

TED (15) - 3111

Reg No:.....

REVISION 2015

Signature:.....

THIRD SEMESTER DIPLOMA EXAMINATION IN TOOL AND DIE ENGINEERING

ENGINEERING METROLOGY

MODEL QUESTION PAPER

[Time: 3 hours

(Maximum marks: 100)

Marks

PART – A

(Maximum marks: 10)

(Answer the following questions in one or two sentences. Each question carries 2 marks)

I

1. Define the term "Accuracy".
2. Write the functions of Ring gauge.
3. Name two types of mechanical comparators.
4. List the parts of a combination set.
5. Define wringing in slip gauge.

(5x2=10)

PART – B

(Maximum marks: 30)

(Answer any *five* of the following questions. Each question carries 6 marks.)

II

1. Explain about direct and indirect method of measurement.

2. Describe about C.L.A method of surface roughness measurement.
3. Explain about the working principle of sine bar.
4. Describe the Machine vision system.
5. State the operation of combination set with sketch.
6. Draw a surface plate and mention three advantages of Glass surface plate.
7. Summerise the procedure to measure angle of V-groove by using balls.

(5x6=30)

PART – C

(Maximum marks - 60)

(Answer one full question from each unit.Each full question carries 15 marks.)

UNIT – I

- III a. Define sesitivity,true value,error,precision. (8)
- b. Sketch and explain the working of a vernier height gauge. (7)

OR

- IV a. Sketch and explain the working principle of micrometer. (8)
- b.Describe about universal surface gauge with sketch. (7)

UNIT – II

- V a,Describe about vernier bevel protractor with sketch. (8)
- b.Explain about the method of finding the angle of Taper hole. (7)

OR

- VI a.Illustrate the internal taper angle measurement of a blind hole steel balls and slip gauges. (8)
- b.Explain about the working principle of Auto-collimator. (7)

UNIT – III

- VII a.Illustrate working procedure of a sigma type mechanical comparator. (8)
- b. Explain about C.L.A and R.M.S method of surface roughness. (7)

OR

- VIII a. Describe about the working of Michelson's interferometer with sketch. (8)
 b. Sketch and explain optical comparator. (7)

UNIT – IV

- IX a. Describe the importance of calibration of measuring instruments. (8)
 b. Explain about the working of Tool maker's microscope with sketch. (7)

OR

- X a. Explain about the working of co-ordinate measuring machine with sketch. (8)
 b. Specify the parts of a vertical optimeter. (7)